

LEARNER GUIDE · CONCEPTS AND COMMAND TYPES

SQL Fundamentals: Relational Databases and Command Types

Understand linked tables, choose the right command family, and explain the differences in an interview.

This guide follows a small student database through structure, queries, permissions and transactions. Start with the relationships, then connect each SQL statement to what it affects.

The running example

Students, Courses and Enrollments show how records connect without repeating every detail.

The memory aid

Structure · Data · Permission · Transaction

Examples use broadly familiar SQL. Syntax and transaction behavior can vary by database system.

RELATIONAL MODEL · TABLES AND KEYS

A relational database stores data in linked tables

Each table describes one kind of thing. Keys connect those things without copying all their details.

A **row** is a record; a **column** is an attribute. A primary key uniquely identifies a row and cannot be null. A foreign key references a key in another table, helping preserve valid relationships.



Physical storage holds the data; tables describe its logical organization.

One student, many enrollments

Table	Key fields	Example record
Students	id (primary key)	1 · Asha · 21
Courses	id (primary key)	10 · SQL Basics
Enrollments	student_id, course_id (foreign keys)	1 · 10

Illustrative records. Enrollments links Students.id to Courses.id; its two fields can form a composite primary key.

Read the relationship

Students → Enrollments and Courses → Enrollments are one-to-many. Together, they implement a many-to-many relationship between students and courses. A one-to-one relationship instead allows at most one matching row on either side.

The **schema** defines tables, columns, types and constraints. MySQL, PostgreSQL, Oracle and SQL Server are examples of relational database systems.

LANGUAGE · SQL AND DATABASE SYSTEMS

SQL is the standard language for talking to them

Structured Query Language is declarative: state what you want, not how to retrieve it.

Example: `SELECT name FROM Students WHERE age > 20;` This asks for names of students older than 20; the engine chooses an execution plan, such as scanning a table or using an available index.

SQL is standardized by ANSI/ISO, with small vendor differences in data types, functions and syntax. Learn the shared ideas, then check the dialect you use.

Why the pairing works

Constraints protect integrity through unique identifiers and valid references; joins combine related records; ACID transactions support atomicity, consistency, isolation and durability.



A client sends SQL to a database engine, which executes it and returns results.

Two interview distinctions

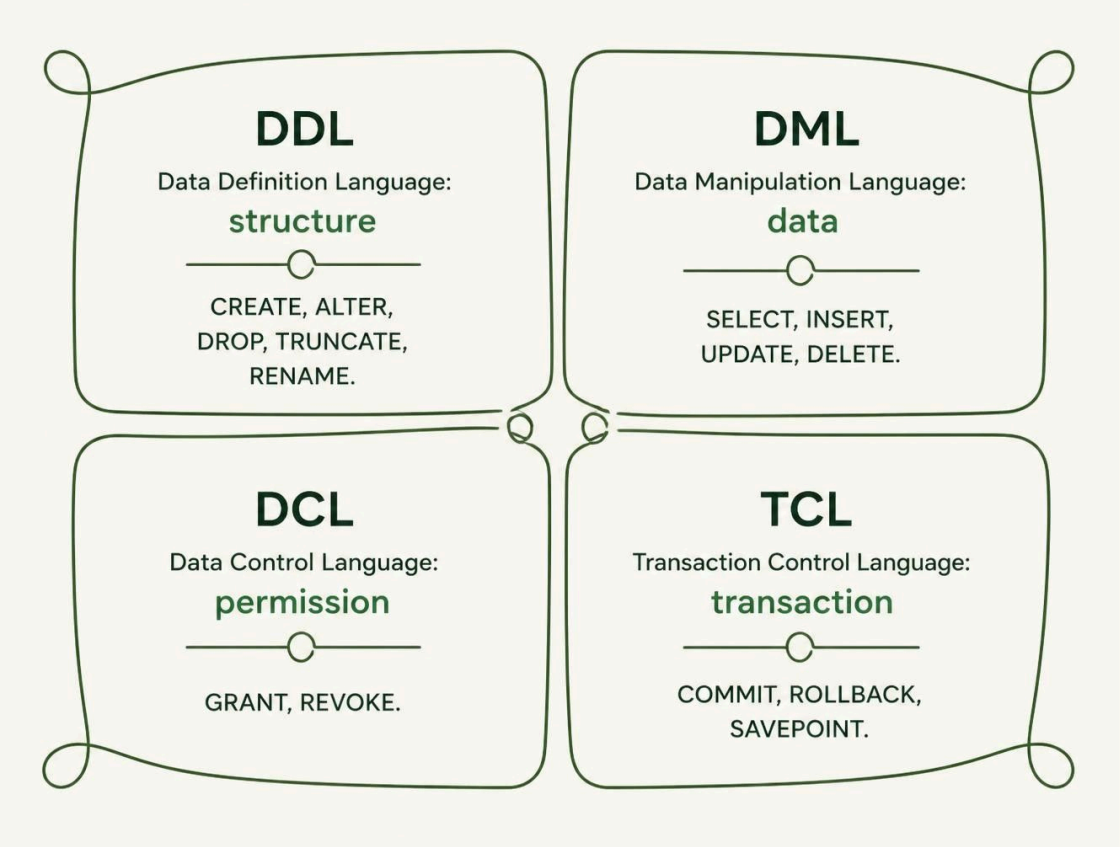
SQL vs MySQL? SQL is a language; MySQL is a database system that implements it.

SQL vs NoSQL? Relational systems organize data in tables; NoSQL covers models such as documents, key-value pairs and graphs. Choose by data model and workload, not by assuming only one category supports transactions.

COMMAND MAP · FOUR FAMILIES

SQL commands fall into four families

Classify a statement by its purpose: structure, data, permission or transaction.



A purpose-based map, not an execution sequence. One application workflow may use several families.

Apply the map to Students

Creating the Students table is DDL. Adding Asha or retrieving her name is DML. Letting an analyst read the table is DCL. Saving or undoing a group of changes is TCL.

Where does SELECT belong?

DQL, or Data Query Language, is sometimes listed separately for SELECT. Many texts place SELECT under DML. This guide uses that convention; explain the convention if an exam or interviewer uses another.

These labels describe intent. They do not, by themselves, tell you whether a command commits automatically or can be rolled back.

DDL · DATA DEFINITION LANGUAGE

DDL defines and changes the structure

DDL creates database objects, changes their definitions or removes them.

Create and alter

Commands: CREATE, ALTER, DROP, TRUNCATE, RENAME.

```
CREATE TABLE Students (
  id INT PRIMARY KEY,
  name VARCHAR(50),
  age INT
);
```

The definition sets column names, data types and a primary-key constraint. It creates the structure, not student records.

```
ALTER TABLE Students
ADD email VARCHAR(100);
```

ALTER changes an existing definition. RENAME changes an object's name; syntax differs by vendor.

Remove or empty

```
DROP TABLE Students;

TRUNCATE TABLE Students;
```

These are alternatives, not statements to run in sequence: after DROP, the table no longer exists.

Command	What it removes	Table remains?
DROP	The table object and its data	No
TRUNCATE	All rows, without WHERE	Yes
DELETE	Matching rows, or all rows	Yes

Interview caution: rollback support, triggers, identity resets and foreign-key restrictions vary by system. DDL does not universally mean “auto-committed.”

DML · DATA MANIPULATION LANGUAGE

DML reads and modifies the data inside tables

INSERT adds records, UPDATE changes them, DELETE removes them, and SELECT retrieves them.

The original Students table has id, name and age; treat DELETE separately if Asha should remain available for the query.

ADD A STUDENT

`INSERT INTO Students VALUES (1, 'Asha', 21);`

Assumes column order; an explicit column list is safer, especially after schema changes.

CHANGE A MATCHING RECORD

`UPDATE Students SET age = 22 WHERE id = 1;`

The primary-key filter targets one student; without WHERE, UPDATE affects every row.

REMOVE A MATCHING RECORD

`DELETE FROM Students WHERE id = 1;`

DELETE removes rows, not the table definition; without WHERE, it removes all rows.

RETRIEVE A FILTERED, SORTED RESULT

`SELECT name FROM Students WHERE age > 20 ORDER BY name;`

SELECT chooses the output column; WHERE filters records; ORDER BY sorts returned names. Reading data does not change stored rows.

Before UPDATE or DELETE, run a SELECT with the same WHERE condition to inspect intended rows. Use an appropriate transaction when changes need review before commit.

DCL AND TCL · ACCESS AND TRANSACTIONS

DCL controls who can do what

GRANT gives privileges; REVOKE takes them back. Assign only the access a user or role needs.

```
GRANT SELECT, INSERT ON Students TO analyst;
```

```
REVOKE INSERT ON Students FROM analyst;
```

Assuming no other grants, analyst can still read Students but can no longer insert rows. Effective access may also come from roles or other grants. User, role and object syntax varies by vendor.

TCL keeps related changes together

A money transfer needs both a debit and a credit to succeed as one unit. This simplified transaction assumes valid accounts, sufficient funds and successful updates.

```
BEGIN;
UPDATE Accounts SET balance = balance - 100 WHERE id = 1;
SAVEPOINT debited;
UPDATE Accounts SET balance = balance + 100 WHERE id = 2;
COMMIT;
```

COMMIT

Makes the transaction's changes permanent.

ROLLBACK

Abandons the uncommitted transaction if the transfer cannot complete.

SAVEPOINT

Marks a point for partial rollback. Rolling back to debited would leave the debit pending, so do not commit an incomplete transfer.

Illustrative transaction syntax. BEGIN and savepoint syntax vary by system. Real transfers also require concurrency controls and validation of affected rows.

DDL, DML, DCL and TCL compared at a glance

Identify what changes first. Then check the database's transaction rules.

Family	Purpose, scope & key commands	One-line example & commit behavior
DDL	Defines structure; affects schema objects. **Key commands:** CREATE, ALTER, DROP, TRUNCATE, RENAME.	`DROP TABLE Students;` Vendor-dependent; some systems implicitly commit DDL.
DML	Reads or changes data; affects rows and query results. **Key commands:** SELECT, INSERT, UPDATE, DELETE.	`SELECT name FROM Students;` Writes follow transaction and autocommit settings; SELECT makes no row changes.
DCL	Controls access; affects privileges. **Key commands:** GRANT, REVOKE.	`GRANT SELECT ON Students TO analyst;` Vendor-dependent; privilege changes may commit implicitly.
TCL	Controls transaction boundaries and pending changes. **Key commands:** COMMIT, ROLLBACK, SAVEPOINT.	`COMMIT;` Explicitly controls transactions; SAVEPOINT does not commit.

Autocommit is a system or connection setting, not a universal property of a command family. SELECT may be classified separately as DQL.

Common interview questions

Primary key vs foreign key? One identifies a row; the other references a key to connect records.

DROP vs TRUNCATE vs DELETE? Remove the object, empty the table, or remove selected rows.

Can every command be rolled back? No universal rule applies. Name the database, command and transaction context.

SQL vs MySQL? A language versus a database system that implements it.